

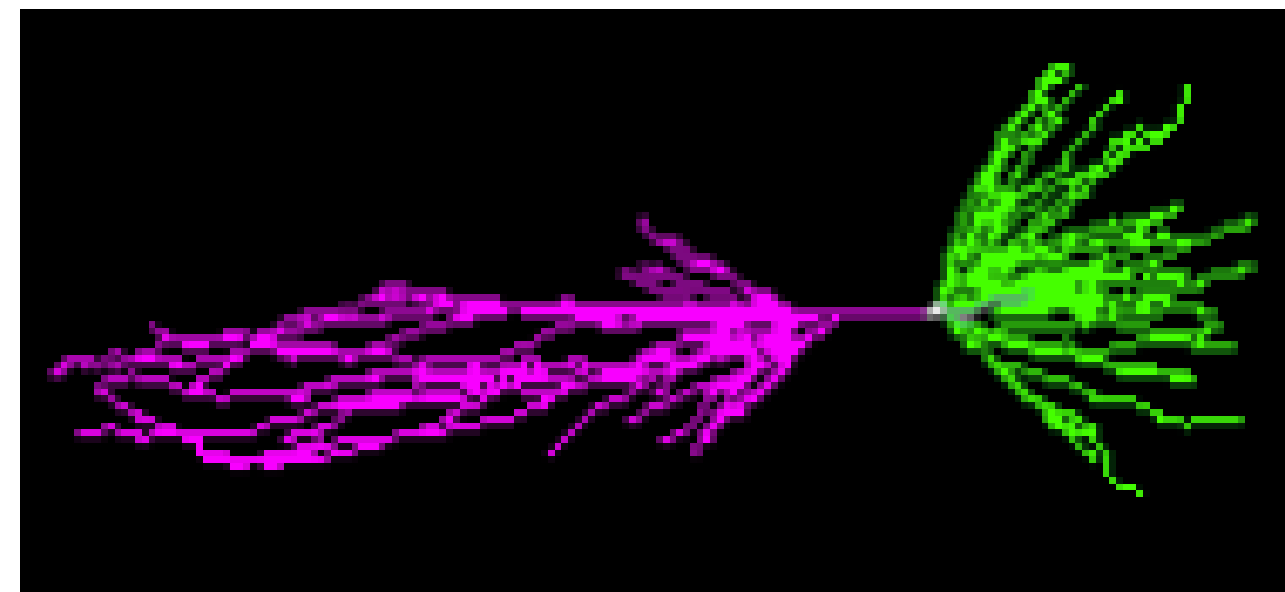
A Surrogate Model for Reducing the Computational Cost of Neuron Simulations

○ Haruki Munechika, Taira Kobayashi

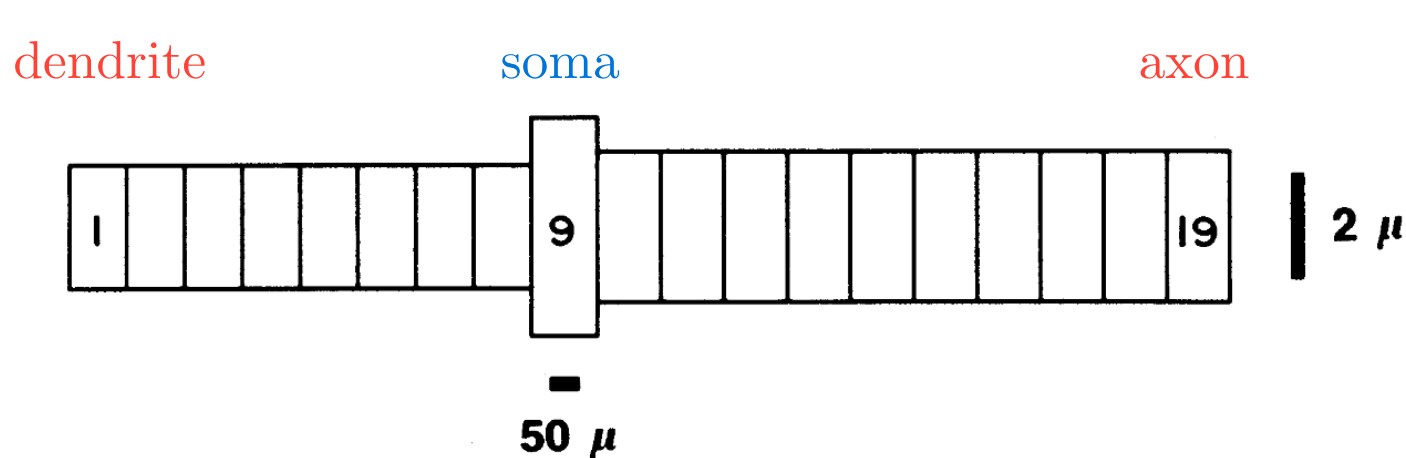
Grad Sch of Sci and Tech for Innov, Yamaguchi University (f032vbw@yamaguchi-u.ac.jp)

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Introduction

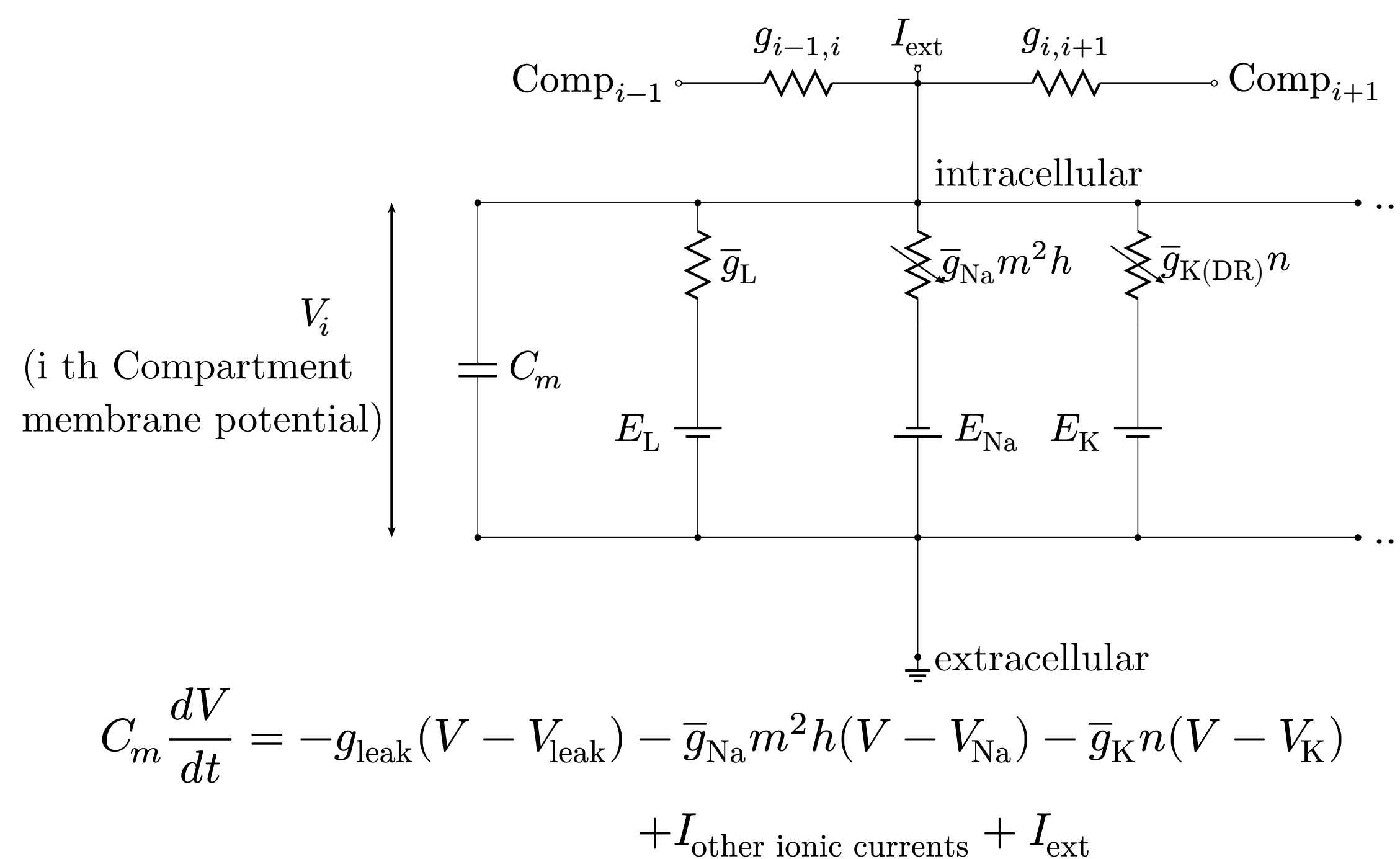


Guinea pig CA3 pyramidal neuron[1]



Modelled as

19 Compartments (comps) [2]



The conductances depend on **gate variables**, which follow ODEs with rate functions α, β :

$$I_{Na} = \bar{g}_{Na} m^2 h (V - E_{Na})$$

$$\frac{dm}{dt} = \alpha_m(V)(1 - m) - \beta_m(V)m$$

$$\alpha_m(V) = 0.32(13.1 - u)/(\exp((13.1 - u)/4) - 1)$$

$$\beta_m(V) = 0.28(u - 40.1)/(\exp((u - 40.1)/5) - 1)$$

→ **11 states (10 gates and V) per comp**

→ **209 states** for 19 comps;

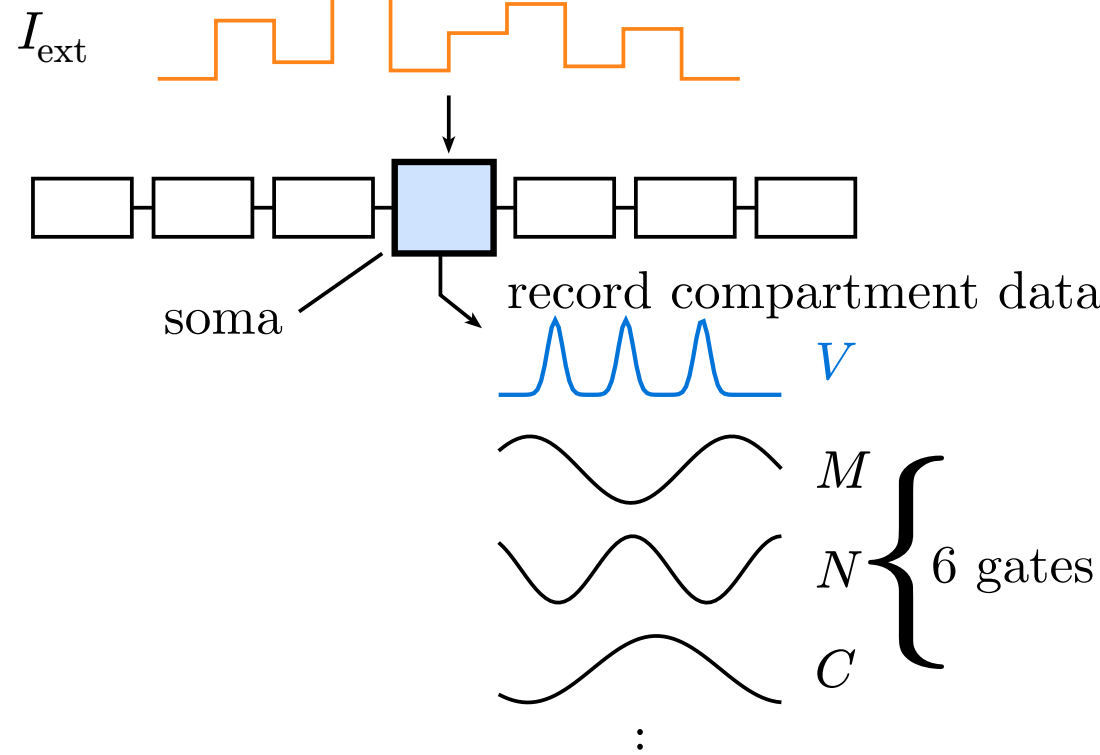
In large scale network simulations, the number of large gate variables becomes a **memory bottleneck**.

Purpose

Development of a surrogate model of the multi-compartment neuron capable of reproducing the membrane potential response with fewer gate variables.

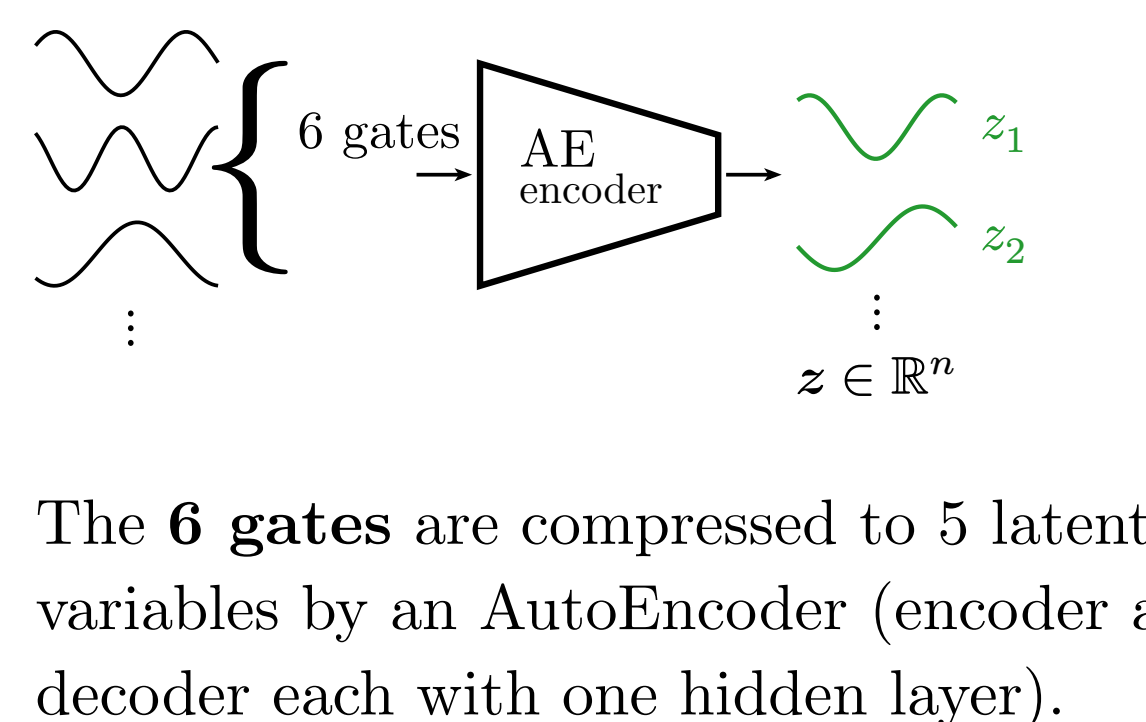
Methods

① Sample the training data



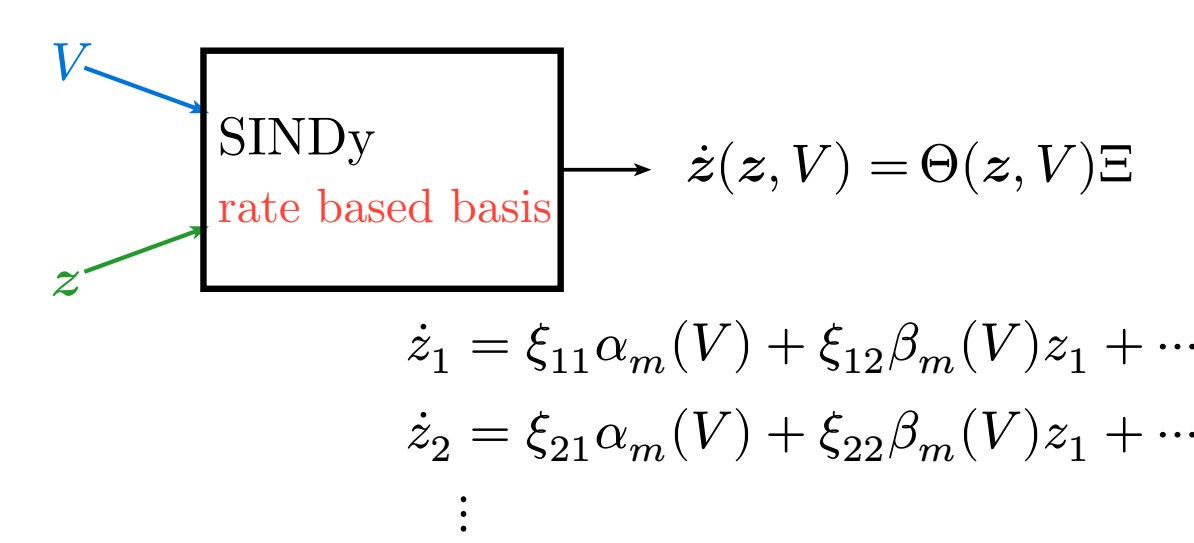
Inject a **random pulse train** at the soma; record V and **6 gates**.

② Compress the gates



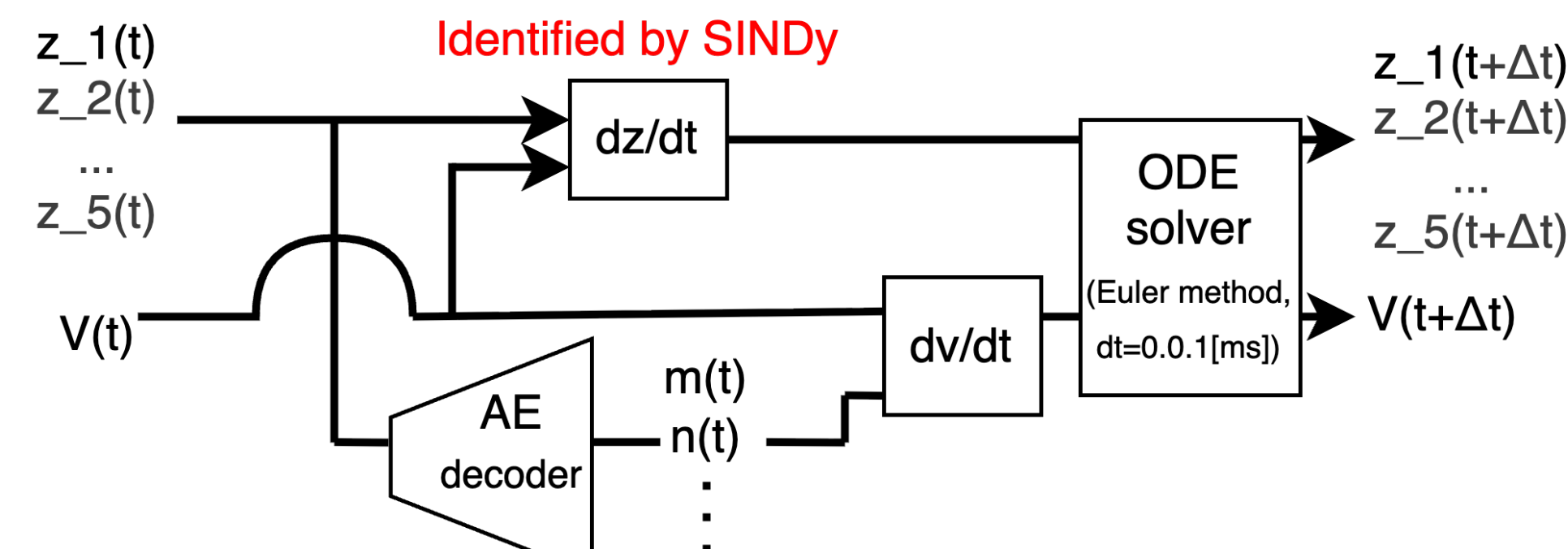
The **6 gates** are compressed to 5 latent variables by an AutoEncoder (encoder and decoder each with one hidden layer).

③ Identify ODEs of the latent variables



Capture the latent variable dynamics with **SINDy**[3]. SINDy fits coefficients Ξ .

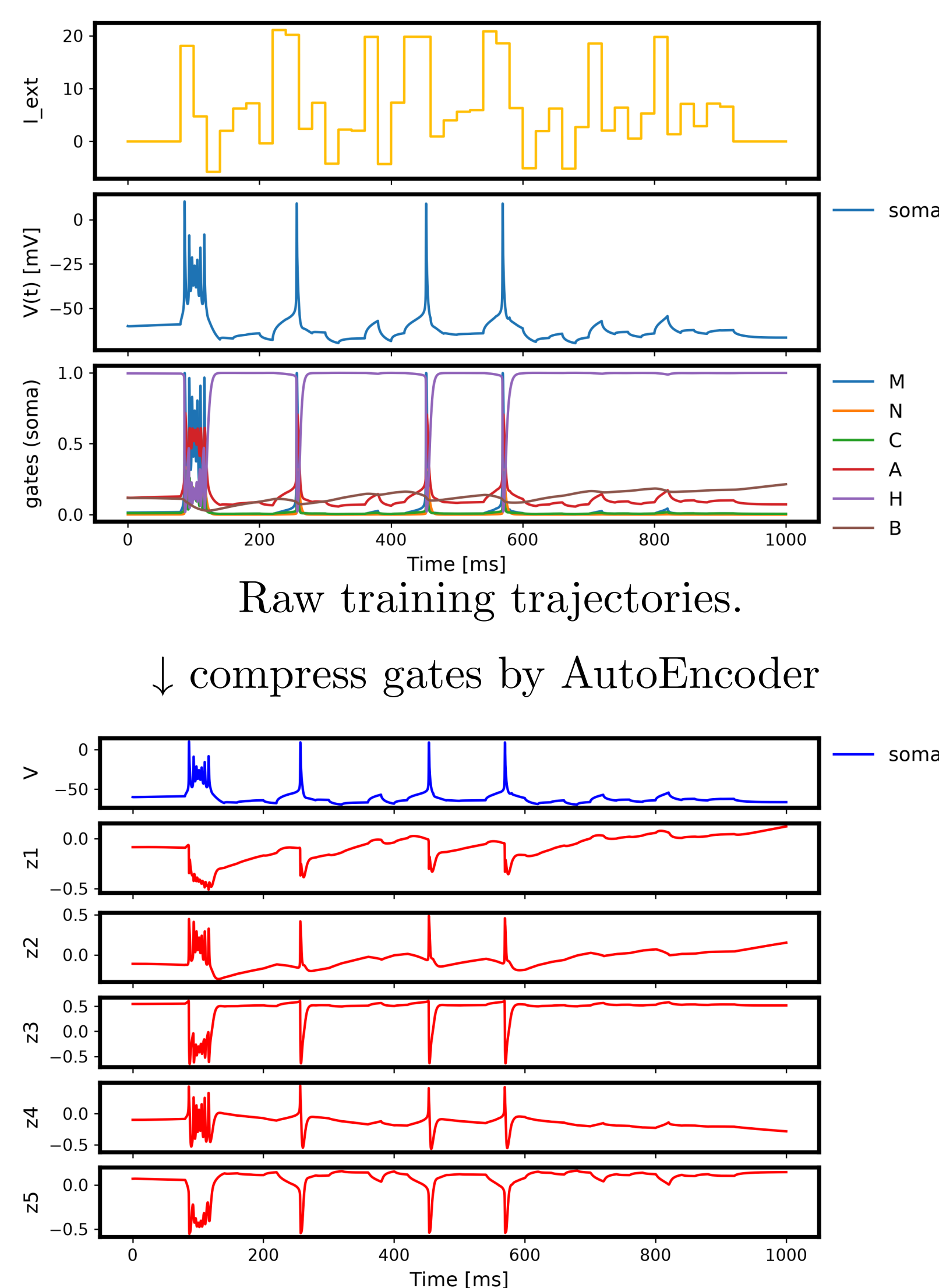
④ How to apply the surrogate model



The derivative of V is computed with the **decoded latent variables**. Across time steps, **only the latent variables and the membrane potential** need to be preserved.

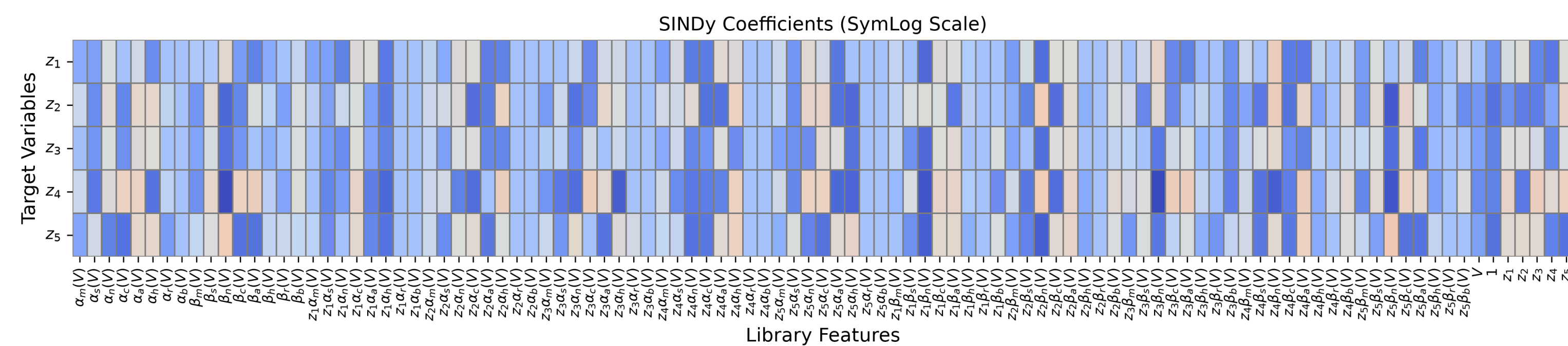
Results and Discussion

Training Data to capture gate dynamics

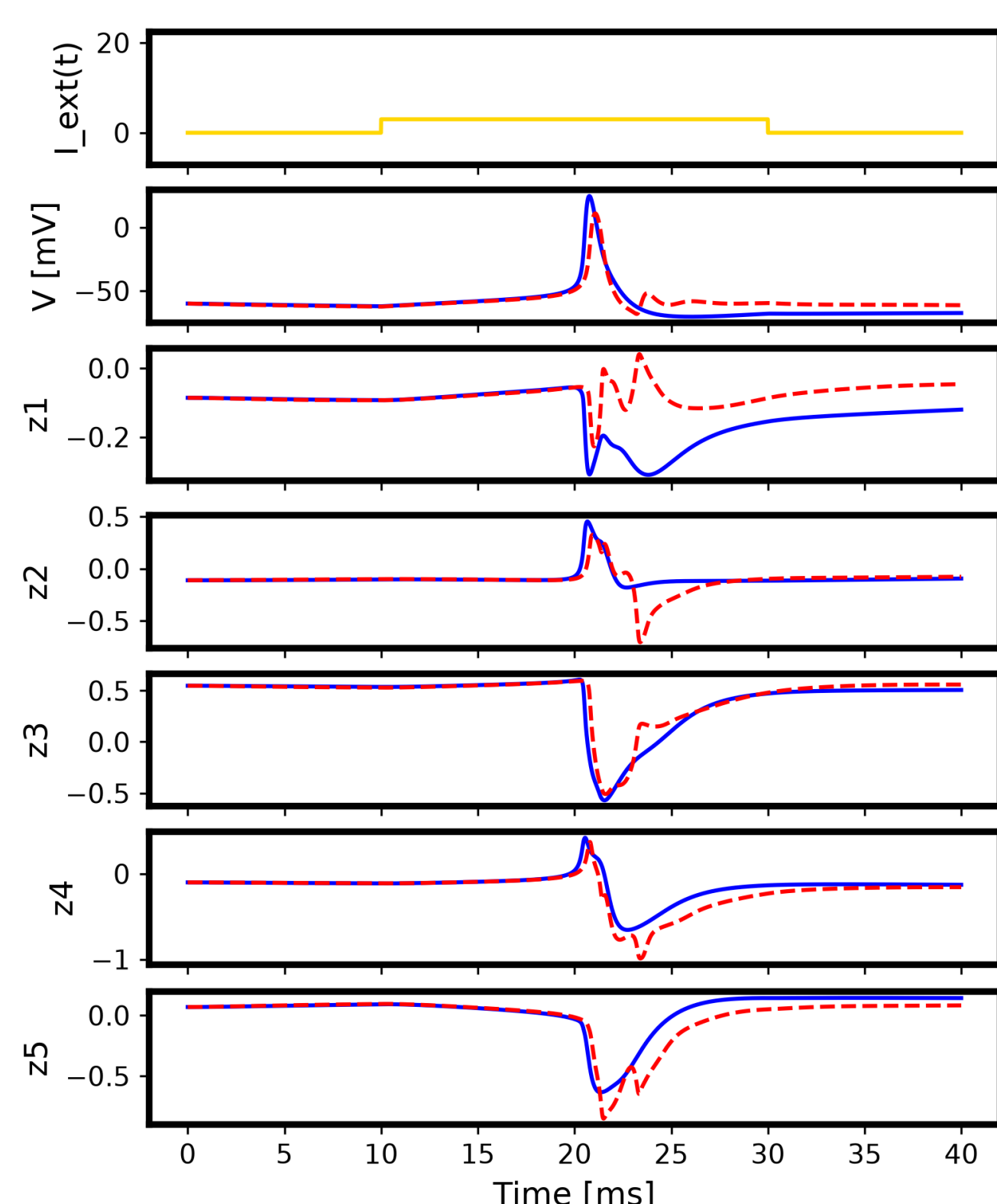


Teaching data to identify the ODEs.

Identified latent equations



Action Potential reproduction

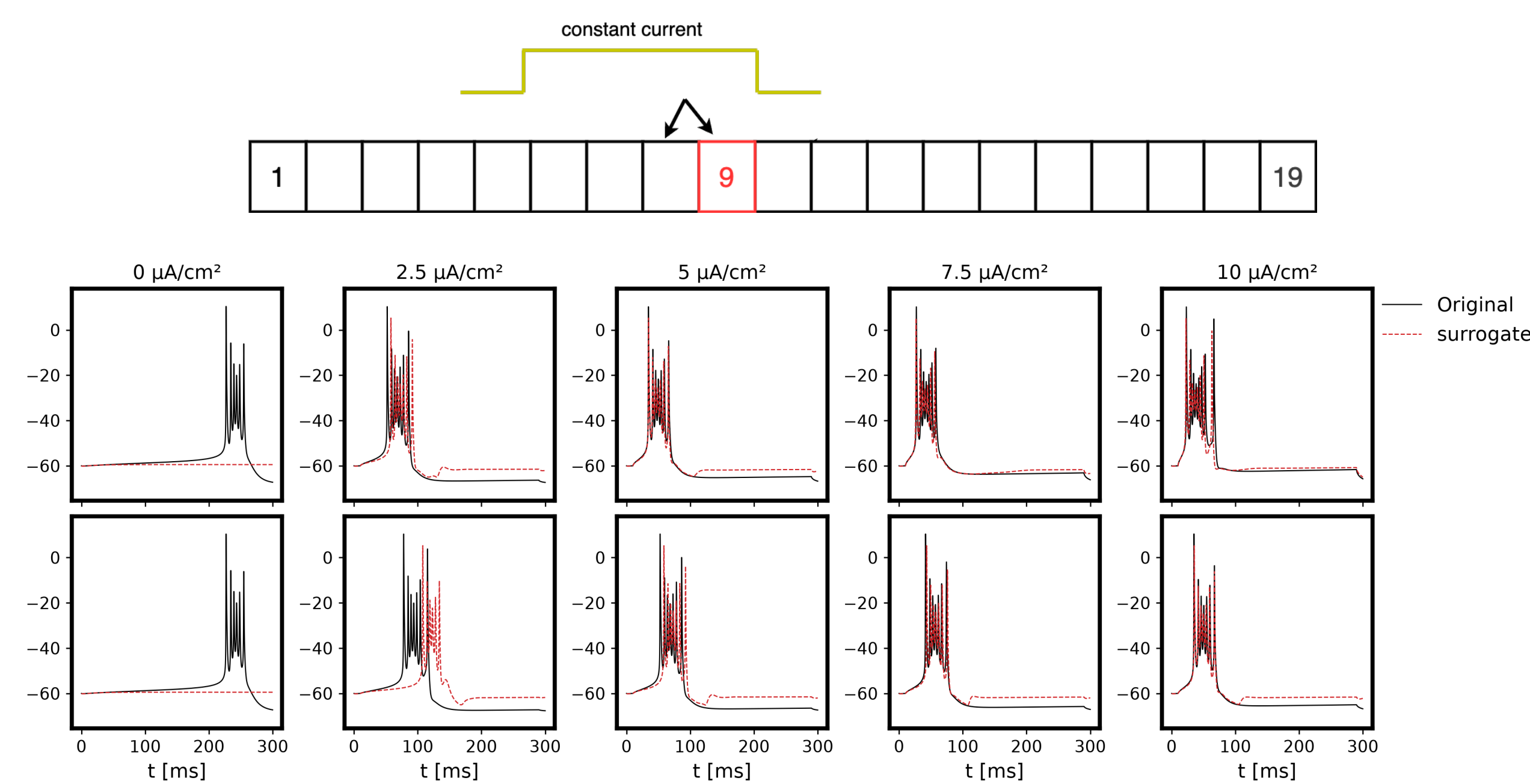


20 ms, 3 μ A/cm² step: V and the **5 AE latents**.

- latency err **0.3 ms**, AHP timing gap **3.1 ms**.
- peak **13 mV** low (amplitude gap **12.9 mV**), rise / fall rate gap **21.0 / 10.3 mV/ms** — yet AHP depth gap is only **0.28 mV**.

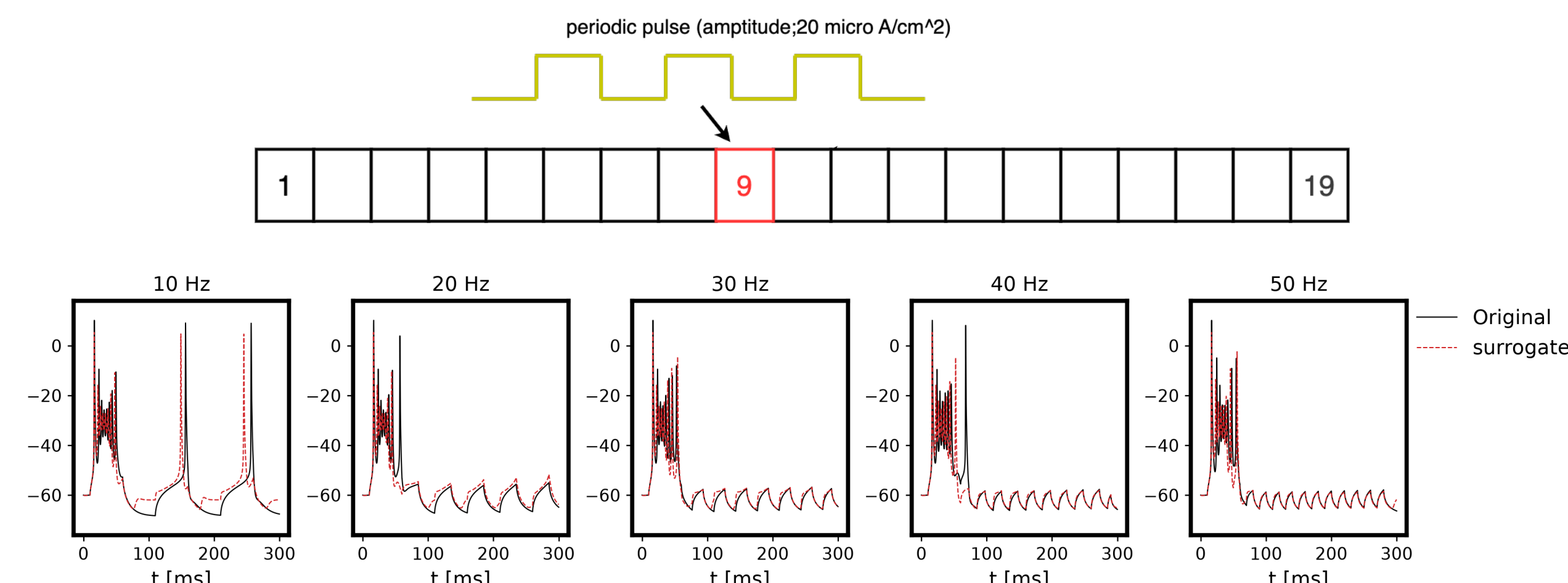
Replace the soma compartment with the surrogate model

soma compartment (9th comp) is replaced with the surrogate model.



Amplitude sweep: **Top**: Inject to soma. **Bottom**: Inject to dendrite.

- The model failed at spontaneous firing but reproduced **bursts** for $I \geq 2.5$: delayed near threshold, and matching spike timing with an elevated post-burst rest for $I \geq 5$.



Frequency sweep: Inject to soma.

- The response to the periodic current is reproduced with high accuracy, despite a slight phase shift.

Conclusion

- Replaced the **soma** compartment only, reducing one gate variable gate variable.
- Waveforms are **accurately reproduced** within the trained stimulus condition.
- Dynamics **outside the training data** like spotaneous firing are **not reproduced**.

Future work

- Explore dimensionality reduction and ODE identification methods that capture the **structure of the original equations** more strongly.
- Replace **all compartments** accurately, with the gate variables reduced further.

Code — github.com/MunechikaHaruki/SINDyNeuroSurrogate

[1] [Online]. Available: https://neuromorpho.org/neuron_info.jsp?neuron_name=L16a

[2] R. D. Traub, R. K. Wong, R. Miles, and H. Michelson, "A Model of a CA3 Hippocampal Pyramidal Neuron Incorporating Voltage-Clamp Data on Intrinsic Conductances," *Journal of Neurophysiology*, vol. 66, no. 2, pp. 635–650, Aug. 1991, doi: 10.1152/jn.1991.66.2.635.

[3] K. Champion, B. Lusch, J. N. Kutz, and S. L. Brunton, "Data-Driven Discovery of Coordinates and Governing Equations," *Proceedings of the National Academy of Sciences*, vol. 116, no. 45, pp. 22445–22451, Nov. 2019, doi: 10.1073/pnas.1906995116.

Conflicts of Interest — The authors declare no conflicts of interest regarding this manuscript.